Title : Threat Intelligence Aggregator (Non-AI)

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Project overview

This project focuses on developing a practical Threat Intelligence (TI) Aggregator designed to collect, parse, normalize, and correlate threat intelligence indicators from multiple external and internal feeds - without using AI or machine learning.

Modern cybersecurity operations rely heavily on accurate and real-time threat intelligence. Indicators of Compromise (IOCs) such as malicious IPs, URLs, domains, and file hashes are distributed across multiple sources, often with different formats and inconsistent structures.

This project helps build a unified system that standardizes these feeds, correlates repeated indicators, and produces blocklists and actionable intelligence reports.

Objectives

- Collect threat intelligence from multiple feeds (files and live sources)
- Parse and extract IOCs (IPs, domains, URLs, hashes, emails)
- Normalize indicators into a unified format
- Correlate indicators across multiple sources
- Generate severity-based threat intelligence
- Create blocklists for security tools
- Produce detailed threat reports

Tools & Technologies

Programming Language:
• Python

Libraries Used:
• re (regex parsing)
• json / csv (data handling)
• requests (live feed fetching)
• ipaddress (IP validation)

Data Sources:
• Local IOC feeds
• Live OSINT feed (Abuse.ch Feodo Tracker)

Key Features

• Multi-source threat intelligence aggregation
• IOC parsing (IP, Domain, URL, Hash, Email)
• Data normalization with metadata
• Correlation engine for repeated indicators
• Severity classification (Low, Medium, High)
• Live threat feed integration
• Automated blocklist generation
• Advanced reporting with top threats

Project Architecture / Workflow

STEP 1: Load Feeds

- Load local files and live threat intelligence feeds

STEP 2: Parse Indicators

- Extract IPs, domains, URLs, hashes, and emails

STEP 3: Normalize Data

- Standardize data format and add metadata

STEP 4: Correlation Engine

- Identify repeated IOCs across feeds

- Assign severity levels

STEP 5: Blocklist Generation

- Generate IP, domain, and URL blocklists

STEP 6: Reporting

- Generate summary report with risk classification and top threats

Output Explanation

The system produces:

• Total number of IOCs processed
• High-risk indicators appearing across multiple feeds
• Top repeated threats
• Categorized IOC lists
• Blocklists for defensive deployment

Screenshots

Figure 1: Script Execution Output

```
>>>
    == RESTART: C:\Users\asus\OneDrive\Desktop\Threat_Intelligence_Project\main.py =
    [INFO] Loading feeds...
    [INFO] Parsing IOCs...
    [INFO] Running correlation engine...
    [INFO] Generating blocklists...
    [INFO] Generating report...

    [✔] Project Completed Successfully!
>>>
```

Figure 2: Parsed IOC Data

```
>>>
    == RESTART: C:/Users/asus/OneDrive/Desktop/Threat_Intelligence_Project/main.py =
    Threat Intelligence Aggregator Started
    IP Feed Data:
     # DstIP
    162.243.103.246
    178.62.3.223
    27.133.154.218
    34.204.119.63
    50.16.16.211
    # END 5 entries
    Domain Feed Data:
     malicious.com
    phishingsite.net
    badactor.org

    URL Feed Data:
     http://malicious.com/malware.exe
    https://phishingsite.net/login

    Extracted IPs: ['162.243.103.246', '178.62.3.223', '27.133.154.218', '34.204.119.63', '50.16.16.211']
    Extracted Domains: ['malicious.com', 'phishingsite.net', 'badactor.org']
    Extracted URLs: ['http://malicious.com/malware.exe', 'https://phishingsite.net/login']
    Valid IPs: ['162.243.103.246', '178.62.3.223', '27.133.154.218', '34.204.119.63', '50.16.16.211']
    Valid Domains: ['malicious.com', 'phishingsite.net', 'badactor.org']
    Valid URLs: ['http://malicious.com/malware.exe', 'https://phishingsite.net/login']
    IOC Database:
    {'indicator': '162.243.103.246', 'type': 'ip', 'source': 'ip_feed', 'timestamp': '2026-03-17 18:37:58.336942'}
    {'indicator': '178.62.3.223', 'type': 'ip', 'source': 'ip_feed', 'timestamp': '2026-03-17 18:37:58.336972'}
    {'indicator': '27.133.154.218', 'type': 'ip', 'source': 'ip_feed', 'timestamp': '2026-03-17 18:37:58.336977'}
    {'indicator': '34.204.119.63', 'type': 'ip', 'source': 'ip_feed', 'timestamp': '2026-03-17 18:37:58.336980'}
    {'indicator': '50.16.16.211', 'type': 'ip', 'source': 'ip_feed', 'timestamp': '2026-03-17 18:37:58.336982'}
    {'indicator': 'malicious.com', 'type': 'domain', 'source': 'domain_feed', 'timestamp': '2026-03-17 18:37:58.336986'}
    {'indicator': 'phishingsite.net', 'type': 'domain', 'source': 'domain_feed', 'timestamp': '2026-03-17 18:37:58.336989'}
    {'indicator': 'badactor.org', 'type': 'domain', 'source': 'domain_feed', 'timestamp': '2026-03-17 18:37:58.336992'}
    {'indicator': 'http://malicious.com/malware.exe', 'type': 'url', 'source': 'url_feed', 'timestamp': '2026-03-17 18:37:58.336996'}
    {'indicator': 'https://phishingsite.net/login', 'type': 'url', 'source': 'url_feed', 'timestamp': '2026-03-17 18:37:58.336999'}
>>>
```

Figure 3: Generated Blocklist

1. IP Blocklist (ip_blocklist.txt)
2. Domain Blocklist (domain_blocklist.txt)
3. URL Blocklist (url_blocklist.txt)

```
IDLE Shell 3.14.3
File  Edit  Shell  Debug  Options  Window  Help
Python 3.14.3 (tags/v3.14.3:323c59a, Feb  3 2026, 16:04:56) [MSC v.1944 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
    == RESTART: C:/Users/asus/OneDrive/Desktop/Threat_Intelligence_Project/main.py =
    Threat Intelligence Aggregator Started
    IP Feed Data:
     # DstIP
    162.243.103.246
    178.62.3.223
    27.133.154.218
    34.204.119.63
    50.16.16.211
    # END 5 entries
    Domain Feed Data:
     malicious.com
    phishingsite.net
    badactor.org

    URL Feed Data:
     http://malicious.com/malware.exe
    https://phishingsite.net/login
```

Figure 4: Threat Intelligence Report

- This is your **final JSON report (MOST IMPORTANT)**

C: > Users > asus > OneDrive > Desktop > Threat_Intelligence_Project > outputs > {} report.json > ...


```
{
    "total_iocs": 10,
    "high_risk": 0,
    "medium_risk": 0,
    "low_risk": 10,
    "top_threats": [
        {
            "indicator": "162.243.103.246",
            "type": "ip",
            "sources": [
                "feed"
            ],
            "count": 1,
            "severity": "LOW"
        },
        {
            "indicator": "178.62.3.223",
            "type": "ip",
            "sources": [
                "feed"
            ],
            "count": 1,
            "severity": "LOW"
        },
        {
            "indicator": "27.133.154.218",
            "type": "ip",
            "sources": [
                "feed"
            ],
            "count": 1,
            "severity": "LOW"
        },
        {
            "indicator": "34.204.119.63",
            "type": "ip",
            "sources": [
```

Practical Applications

- SOC threat monitoring
- Firewall and IDS/IPS blocklist generation
- Malware infrastructure detection
- Threat intelligence automation

Learning Outcomes

- Understanding of threat intelligence feeds
- IOC parsing and validation techniques
- Data normalization and correlation
- SOC workflow simulation
- Automation in cybersecurity operations

Future Enhancements

- Integration with VirusTotal API
- Support for STIX/TAXII feeds
- Real-time alerting system
- Dashboard visualization
- Machine learning-based threat scoring

Conclusion

This project demonstrates a practical implementation of threat intelligence aggregation and analysis. It highlights key SOC processes such as IOC parsing, correlation, and proactive defense through blocklist generation.

The system provides a strong foundation for further development into advanced cybersecurity tools.